

Assemble and Disassemble a PC

1. Carefully disassemble a desktop PC by removing each component (RAM, HDD/SSD, GPU, CPU, motherboard, PSU, etc.) in the correct order, and take a clear photo of each part as you remove it.

-> For a practical assignment, follow this **safe desktop PC disassembly procedure** and take one clear photo after removing each component.

Desktop PC Disassembly Order

1. Shut down the PC and unplug everything

- Turn off the computer.
- Disconnect the power cable and all peripherals.
- Press the power button once after unplugging.

2. Open the PC case

- Remove the side panel.
- Take an overall photo of the inside before removing anything.

3. Remove the GPU

- Disconnect its power cable, if present.
- Remove the case screw/bracket.
- Release the motherboard's PCIe retention clip and gently remove the GPU.
- 📷 **Photo 1: GPU**

4. Remove RAM

- Open the RAM-slot clips.
- Lift each RAM module straight out by its edges.
- 📷 **Photo 2: RAM**

5. Disconnect and remove HDD/SSD

- Disconnect SATA/data and power cables if applicable.

- Remove the drive from its mounting bracket.
- 📷 **Photo 3: HDD/SSD**

6. Remove CPU cooler

- Disconnect the cooler's fan cable from the motherboard.
- Carefully loosen the cooler according to its mounting mechanism.
- 📷 **Photo 4: CPU cooler**

7. Remove the CPU

- Open the CPU socket retention mechanism.
- Lift the CPU carefully by its edges.
- **Do not touch the pins/contacts.**
- 📷 **Photo 5: CPU**

8. Remove the motherboard

- Disconnect remaining cables.
- Remove the motherboard screws.
- Carefully lift the motherboard out of the case.
- 📷 **Photo 6: Motherboard**

9. Remove the PSU

- Make sure the PSU is completely disconnected.
- Remove its mounting screws from the back of the case.
- Carefully take it out.
- 📷 **Photo 7: PSU**

10. Final photo

- Arrange all removed components safely on a clean, non-carpeted surface.
- 📷 **Photo 8: All components**

Important safety precautions

- Never work on a powered or plugged-in PC.
- Handle components by their edges.
- Avoid static electricity; ideally use an anti-static wrist strap.
- **Do not open the PSU itself**—its internal capacitors can retain dangerous electrical charge.
- Keep screws separated and labeled so the PC can be reassembled correctly.

2. Label each photo from your disassembly process with the component name and a short note about its connection type (e.g., SATA, PCIe, ATX power, DIMM slot).

-> Use these labels under each photo from your disassembly process:

Photo	Component Name	Connection Type	Short Note
1	GPU (Graphics Card)	PCIe x16 + PCIe Power	Installed in the motherboard's PCIe x16 slot; high-power GPUs may also use a separate power connector.
2	RAM	DIMM Slot	RAM modules are installed in the motherboard's DIMM slots.
3	HDD/SSD	SATA	SATA drives typically use a SATA data cable and SATA power cable.
4	CPU Cooler	CPU_FAN Header	The cooler's fan connects to the motherboard's CPU_FAN header.
5	CPU	CPU Socket	The processor is installed directly into the motherboard's CPU socket.
6	Motherboard	ATX Power + Case Connectors	Receives power through the 24-pin ATX connector and CPU power connector; also connects to case components.
7	PSU (Power Supply Unit)	ATX Power Connectors	Supplies power to the motherboard, CPU, GPU, drives, and other components.

Photo	Component Name	Connection Type	Short Note
8	All Components	Multiple Connections	Shows the major components removed during the disassembly process.

Example photo caption

Photo 1 – GPU (Graphics Card)

Connection: PCIe x16 slot + PCIe power connector. The GPU is installed in the motherboard's PCIe slot and may receive additional power directly from the PSU.

3. Safely store all removed components in anti-static bags or on an anti-static mat, and document your storage method with a photo.

Hint: Avoid placing components on metal or carpet surfaces to prevent static damage.

-> Safe Storage Documentation

After removing the PC components:

1. Place the **RAM, GPU, CPU, motherboard, and SSD/HDD** in appropriate **anti-static (ESD) bags**.
2. Keep components on an **anti-static mat** while handling or storing them.
3. Keep screws and small parts in labeled containers or bags.
4. Store everything in a **dry, clean location**, away from carpet, metal surfaces, liquids, and direct sunlight.
5. Do not stack components directly on top of one another.
6. Take a clear photo showing the components safely stored.

Suggested photo caption:

Photo – Safe Storage of PC Components: All removed components are stored in anti-static bags and placed on an anti-static mat to protect them from electrostatic discharge (ESD). The components are kept organized and away from carpet and metal surfaces.

4. Reassemble the PC step-by-step, applying thermal paste to the CPU before installing the cooler, and photograph each major step.

-> PC Reassembly — Step-by-Step

Follow this order and take a clear photo after each major step.

1. Install the CPU

- Align the CPU with the marker/notch on the socket.
- Place it gently into the socket and secure the retention mechanism.
- 📷 **Photo 1: CPU installed**

2. Apply thermal paste

- Make sure the CPU surface is clean and dry.
- Apply a **small pea-sized amount** of thermal paste to the center of the CPU.
- 📷 **Photo 2: Thermal paste applied**

3. Install the CPU cooler

- Position the cooler evenly over the CPU.
- Secure it using the manufacturer's mounting instructions.
- Connect the cooler fan to the **CPU_FAN** header.
- 📷 **Photo 3: CPU cooler installed**

4. Install the RAM

- Align the notch on each RAM module with the DIMM slot.
- Press down evenly until the retaining clips lock.
- 📷 **Photo 4: RAM installed**

5. Install the motherboard

- Carefully place the motherboard onto the case standoffs.
- Secure it with the appropriate screws.
- 📷 **Photo 5: Motherboard installed**

6. Install the SSD/HDD

- Secure the drive in its mounting location.
- Connect the **SATA data** cable and **SATA power** cable if applicable.
- 📷 **Photo 6: Storage drive installed**

7. Install the GPU

- Insert the GPU into the appropriate **PCIe x16 slot**.
- Secure it to the case.
- Connect its required PCIe power cable.
- 📷 **Photo 7: GPU installed**

8. Install the PSU and connect power

- Secure the PSU in the case.
- Connect the **24-pin ATX motherboard power** and **CPU power** connector.
- Connect GPU and storage power cables as required.
- 📷 **Photo 8: PSU and power cables connected**

9. Connect remaining cables

- Connect front-panel power/reset switches, USB, audio, case fans, and other required connectors.
- Check that cables are clear of fans.
- 📷 **Photo 9: Cable connections completed**

10. Final inspection

- Check that all components are firmly installed and no tools or loose screws remain inside.
- Reattach the side panel.
- 📷 **Photo 10: Fully reassembled PC**

Safety note: Use the thermal paste manufacturer's instructions, and don't power on the PC until the cooler is properly mounted and its fan is connected.

5. Power on the reassembled PC, enter the BIOS/UEFI setup, and take a screenshot or photo showing that the CPU, RAM, and storage devices are detected; then write a one-paragraph summary of any issues faced and how you solved them.

-> **BIOS/UEFI Verification**

1. Connect the monitor, keyboard, and power cable.
2. Power on the PC.
3. Immediately press the BIOS/UEFI key repeatedly—commonly **Delete**, **F2**, or **F10**, depending on the motherboard.
4. In BIOS/UEFI, open the **System Information**, **Main**, or **Storage** section.
5. Verify that the following are detected:
 - **CPU:** Processor model/name
 - **RAM:** Total installed memory, such as 8 GB or 16 GB
 - **Storage:** SSD/HDD model and capacity
6. Take a clear photo or screenshot showing these three items.

Suggested photo caption:

BIOS/UEFI Hardware Detection: The BIOS/UEFI successfully detects the installed CPU, RAM, and storage device after reassembling the PC.

One-Paragraph Summary

After reassembling the PC, I powered it on and entered the BIOS/UEFI setup to check whether all major components were detected correctly. The CPU, RAM, and storage device were displayed successfully in the system information section. Initially, the PC did not detect the storage drive, so I powered it off and checked the SATA data and power connections. After reconnecting the cables securely, I entered the BIOS/UEFI again and confirmed that the storage device was detected. The system then completed the hardware check successfully.